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Seamless gapfill of metal nitrides

Abstract

Methods for filling a substrate feature with a seamless metal gate fill are described. Methods comprise sequentially depositing a film on a substrate surface having at least one feature thereon. The at least one feature extends a feature depth from the substrate surface to a bottom surface and has a width defined by a first sidewall and a second sidewall. The film is treated with an oxidizing plasma. Then the film is etched to remove the oxidized film. A second film is deposited to fill the feature, where the second film substantially free of seams and voids.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Application No. 63/208,338, filed Jun. 8, 2021, and U.S. Provisional Application No. 63/217,492, filed Jul. 1, 2021, the entire disclosures of which are hereby incorporated by reference herein.

TECHNICAL FIELD
(1) Embodiments of the disclosure generally relate to methods for filling substrate features. More particularly, embodiments of the disclosure are directed to methods for filling a substrate feature with a deposition-treatment-etch-deposition scheme.
BACKGROUND
(2) Integrated circuits have evolved into complex devices that can include millions of transistors, capacitors, and resistors on a single chip. In the course of integrated circuit evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased.
(3) The transistor is a key component of most integrated circuits. Since the drive current, and therefore speed, of a transistor is proportional to the gate width of the transistor, faster transistors

generally require larger gate width. Thus, there is a trade-off between transistor size and speed, and “fin” field-effect transistors (finFETs) have been developed to address the conflicting goals of a transistor having maximum drive current and minimum size. FinFETs are characterized by a fin-shaped channel region that greatly increases the size of the transistor without significantly increasing the footprint of the transistor and are now being applied in many integrated circuits. Examples of transistor device structures include a planar structure, a fin field effect transistor (FinFET) structure, and a gate all around (GAA) structure.
(4) In manufacturing, three-dimensional (3D) structures, such as FinFETs, GAAs, DRAM word lines, and the like, are made by atomic layer deposition of titanium nitride (TiN) and tungsten (W) stacks. A seam is often observed from an ALD or CVD deposited film. This seam creates issues during downstream processes. Accordingly, there is a need for methods to fill structures without creating a seam.
SUMMARY
(5) One or more embodiments of the disclosure are directed to a processing method. The processing method comprises depositing a first film on a substrate surface, the substrate surface having at least one feature thereon, the at least one feature extending a feature depth from the substrate surface to a bottom surface, the at least one feature having a width defined by a first sidewall and a second sidewall; treating the substrate surface with an oxidizing plasma to form an oxidized film; etching the substrate surface to remove the oxidized film; and depositing a second film on the substrate surface to fill the at least one feature, the second film substantially free of seams and voids.
(6) Another embodiment of the disclosure is directed to a processing method. The processing method comprising: forming a film stack on a substrate, the film stack comprising a plurality of alternating layers of a first material and a second material and the film stack having a stack thickness; etching the film stack to form an opening extending a depth from a top of the film stack surface to a bottom surface, the opening having a width defined by a first sidewall and a second sidewall; depositing a first film on the film stack surface, and on the first sidewall, the second sidewall, and the bottom surface of the opening; treating the film stack with an oxidizing plasma to form an oxidized film on the film stack surface; etching the oxidized film to expose the first film; depositing a second film on the film stack surface, the second film substantially free of seams and voids.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.
(2) FIG. **1** shows a cross-sectional isometric view of a processing chamber in accordance with one or more embodiments of the disclosure;
(3) FIG. **2** shows a cross-sectional view of a processing chamber in accordance with one or more embodiments of the disclosure;
(4) FIG. **3** is an exploded cross-sectional view of a processing station in accordance with one or more embodiments of the disclosure;
(5) FIG. **4** is a schematic representation of a processing platform in accordance with one or more embodiments of the disclosure;

(6) FIG. **5** shows a cross-sectional view of a substrate feature in accordance with one or more embodiments of the disclosure;
(7) FIGS. **6**A-**6**D show a cross-sectional view of a substrate in accordance with the method of one or more embodiments of the disclosure; and
(8) FIG. **7** shows a process flow in accordance with one or more embodiment of the disclosure.

DETAILED DESCRIPTION

(9) Before describing several exemplary embodiments of the disclosure, it is to be understood that the disclosure is not limited to the details of construction or process steps set forth in the following description. The disclosure is capable of other embodiments and of being practiced or being carried out in various ways.

(10) As used in this specification and the appended claims, the term “substrate” and “wafer” are used interchangeably, both referring to a surface, or portion of a surface, upon which a process acts. It will also be understood by those skilled in the art that reference to a substrate can also refer to only a portion of the substrate unless the context clearly indicates otherwise. Additionally, reference to depositing on a substrate can mean both a bare substrate and a substrate with one or more films or features deposited or formed thereon.

(11) A “substrate” as used herein, refers to any substrate or material surface formed on a substrate upon which film processing is performed during a fabrication process. For example, a substrate surface on which processing can be performed include materials such as silicon, silicon oxide, strained silicon, silicon on insulator (SOI), carbon doped silicon oxides, silicon nitride, doped silicon, germanium, gallium arsenide, glass, sapphire, and any other materials such as metals, metal nitrides, metal alloys, and other conductive materials, depending on the application. Substrates include, without limitation, semiconductor wafers. Substrates may be exposed to a pretreatment process to polish, etch, reduce, oxidize, hydroxylate (or otherwise generate or graft target chemical moieties to impart chemical functionality), anneal and/or bake the substrate surface. In addition to film processing directly on the surface of the substrate itself, in the present disclosure, any of the film processing steps disclosed may also be performed on an underlayer formed on the substrate as disclosed in more detail below, and the term “substrate surface” is intended to include such underlayer as the context indicates. Thus, for example, where a film/layer or partial film/layer has been deposited onto a substrate surface, the exposed surface of the newly deposited film/layer becomes the substrate surface. What a given substrate surface comprises will depend on what films are to be deposited, as well as the particular chemistry used.

(12) As used in this specification and the appended claims, the terms “precursor”, “reactant”, “reactive gas” and the like are used interchangeably to refer to any gaseous species that can react with the substrate surface.

(13) According to one or more embodiments, the term “on”, with respect to a film or a layer of a film, includes the film or layer being directly on a surface, for example, a substrate surface, as well as there being one or more underlayers between the film or layer and the surface, for example the substrate surface. Thus, in one or more embodiments, the phrase “on the substrate surface” is intended to include one or more underlayers. In other embodiments, the phrase “directly on” refers to a layer or a film that is in contact with a surface, for example, a substrate surface, with no intervening layers. Thus, the phrase “a layer directly on the substrate surface” refers to a layer in direct contact with the substrate surface with no layers in between.

(14) Embodiments of the disclosure provide methods of depositing a metal nitride film (e.g., titanium nitride (TiN), or silicon nitride (SiN)) in high aspect ratio (AR) structures with small dimensions. Some embodiments advantageously provide methods involving cyclic deposition-treatment-etch-deposition processes that can be performed in a batch processing chamber. Some embodiments advantageously provide seam-free high-quality films to fill up high AR trenches with small dimensions.

(15) The disclosure provides methods for use with single wafer or multi-wafer (also referred to as

batch) process chambers. FIGS. **1** and **2** illustrate a processing chamber **100** in accordance with one or more embodiment of the disclosure. FIG. **1** shows the processing chamber **100** illustrated as a cross-sectional isometric view in accordance with one or more embodiment of the disclosure. FIG. **2** shows a processing chamber **100** in cross-section according to one or more embodiment of the disclosure. Accordingly, some embodiments of the disclosure are directed to processing chambers **100** that incorporate a substrate support **200**.
(16) The processing chamber **100** has a housing **102** with walls **104** and a bottom **106**. The housing **102** along with the top plate **300** define a processing volume **109**, also referred to as an interior volume.
(17) The processing chamber **100** illustrated includes a plurality of processing stations **110**. The processing stations **110** are located in the interior volume **109** of the housing **102** and are positioned in a circular arrangement around the rotational axis **211** of the substrate support **200**. Each processing station **110** comprises a gas distribution plate **112** (also referred to as a gas injector) having a front surface **114**. In some embodiments, the front surfaces **114** of each of the gas injectors **112** are substantially coplanar. The processing stations **110** are defined as a region in which processing can occur. For example, in some embodiments, a processing station **110** is defined as a region bounded by the support surface **231** of the substrate support **200**, as described below, and the front surface **114** of the gas injectors **112**. In the illustrated embodiment, heaters **230** act as the substrate support surfaces and form part of the substrate support **200**.
(18) The processing stations **110** can be configured to perform any suitable process and provide any suitable process conditions. The type of gas distribution plate **112** used will depend on, for example, the type of process being performed and the type of showerhead or gas injector. For example, a processing station **110** configured to operate as an atomic layer deposition apparatus may have a showerhead or vortex type gas injector. Whereas a processing station **110** configured to operate as a plasma station may have one or more electrode and/or grounded plate configuration to generate a plasma while allowing a plasma gas to flow toward the wafer. The embodiment illustrated in FIG. **2** has a different type of processing station **110** on the left side (processing station **110***a*) of the drawing than on the right side (processing station **110***b*) of the drawing. Suitable processing stations **110** include, but are not limited to, thermal processing stations, microwave plasma, three-electrode CCP, ICP, parallel plate CCP, UV exposure, laser processing, pumping chambers, annealing stations and metrology stations.
(19) FIG. **3** illustrates an exploded view of a gas distribution assembly **105** for use in a processing station **110** or a process chamber in accordance with one or more embodiment of the disclosure. The skilled artisan will recognize that embodiment illustrated in FIG. **3** is a general schematic and omits details (e.g., gas channels). The gas distribution assembly **105** illustrated comprises three main components: a gas distribution plate **112**, a lid **180** and an optional spacer **330**. The spacer **330** is also referred to as a pump/purge spacer, insert or pump/purge insert. In some embodiments, the spacer **330** is connected to or in fluid communication with a vacuum (exhaust). In some embodiments, the spacer **330** is connected to or in fluid communication with a purge gas source.
(20) The openings **310** in the top plate **300** can be uniformly sized or have different sizes. Different sized/shape gas injectors **112** can be used with a pump/purge spacer **330** that is suitably shaped to transition from the opening **310** to the gas distribution plate **112**. For example, as illustrated, the pump/purge spacer **330** includes a top **331** and bottom **333** with a sidewall **335**. When inserted into the opening **310** in the top plate **300**, a ledge **334** is configured to be positioned in the opening **310**.
(21) The pump/purge spacer **330** includes an opening **339** in which a gas distribution plate **112** can be inserted. The gas distribution plate **112** illustrated has a flange **342** which can be in contact with the ledge formed by the back surface **332** adjacent the top **331** of the pump/purge spacer **330**. The diameter or width of the gas distribution plate **112** can be any suitable size that can fit within the opening **339** of the pump/purge spacer **330**. This allows gas injectors **112** of various types to be used within the same opening **310** in the top plate **300**.

(22) FIG. **4** shows a processing platform **400** in accordance with one or more embodiment of the disclosure. The embodiment shown in FIG. **4** is merely representative of one possible configuration and should not be taken as limiting the scope of the disclosure. For example, in some embodiments, the processing platform **400** has a different number of one or more of the processing chambers **100**, buffer stations **420** and/or robot **430** configurations than the illustrated embodiment.
(23) The exemplary processing platform **400** includes a central transfer station **410** which has a plurality of sides **411**, **412**, **413**, **414**. The transfer station **410** shown has a first side **411**, a second side **412**, a third side **413** and a fourth side **414**. Although four sides are shown, those skilled in the art will understand that there can be any suitable number of sides to the transfer station **410** depending on, for example, the overall configuration of the processing platform **400**. In some embodiments, there the transfer station **410** has three sides, four sides, five sides, six sides, seven sides or eight sides.
(24) The transfer station **410** has a robot **430** positioned therein. The robot **430** can be any suitable robot capable of moving a wafer during processing. In some embodiments, the robot **430** has a first arm **431** and a second arm **432**. The first arm **431** and second arm **432** can be moved independently of the other arm. The first arm **431** and second arm **432** can move in the x-y plane and/or along the z-axis. In some embodiments, the robot **430** includes a third arm (not shown) or a fourth arm (not shown). Each of the arms can move independently of other arms.
(25) The embodiment illustrated includes six processing chambers **100** with two connected to each of the second side **412**, third side **413** and fourth side **414** of the central transfer station **410**. Each of the processing chambers **100** can be configured to perform different processes.
(26) The processing platform **400** can also include one or more buffer station **420** connected to the first side **411** of the central transfer station **410**. The buffer stations **420** can perform the same or different functions. For example, the buffer stations may hold a cassette of wafers which are processed and returned to the original cassette, or one of the buffer stations may hold unprocessed wafers which are moved to the other buffer station after processing. In some embodiments, one or more of the buffer stations are configured to pre-treat, pre-heat or clean the wafers before and/or after processing.
(27) The processing platform **400** may also include one or more slit valves **418** between the central transfer station **410** and any of the processing chambers **100**. The slit valves **418** can open and close to isolate the interior volume within the processing chamber **100** from the environment within the central transfer station **410**. For example, if the processing chamber will generate plasma during processing, it may be helpful to close the slit valve for that processing chamber to prevent stray plasma from damaging the robot in the transfer station.
(28) The processing platform **400** can be connected to a factory interface **450** to allow wafers or cassettes of wafers to be loaded into the processing platform **400**. A robot **455** within the factory interface **450** can be used to move the wafers or cassettes into and out of the buffer stations. The wafers or cassettes can be moved within the processing platform **400** by the robot **430** in the central transfer station **410**. In some embodiments, the factory interface **450** is a transfer station of another cluster tool (i.e., another multiple chamber processing platform).
(29) A controller **495** may be provided and coupled to various components of the processing platform **400** to control the operation thereof. The controller **495** can be a single controller that controls the entire processing platform **400**, or multiple controllers that control individual portions of the processing platform **400**. For example, the processing platform **400** of some embodiments comprises separate controllers for one or more of the individual processing chambers **100**, central transfer station **410**, factory interface **450** and/or robots **430**.
(30) In some embodiments, the processing chamber **100** further comprises a controller **495** connected to the plurality of substantially coplanar support surfaces **231** configured to control one or more of the first temperature or the second temperature. In one or more embodiments, the controller **495** controls a movement speed of the substrate support **200** (FIG. **2**).

(31) In some embodiments, the controller **495** includes a central processing unit (CPU) **496**, a memory **497**, and support circuits **498**. The controller **495** may control the processing platform **400** directly, or via computers (or controllers) associated with particular process chamber and/or support system components.
(32) The controller **495** may be one of any form of general-purpose computer processor that can be used in an industrial setting for controlling various chambers and sub-processors. The memory **497** or computer readable medium of the controller **495** may be one or more of readily available memory such as random access memory (RAM), read only memory (ROM), floppy disk, hard disk, optical storage media (e.g., compact disc or digital video disc), flash drive, or any other form of digital storage, local or remote. The memory **497** can retain an instruction set that is operable by the processor (CPU **496**) to control parameters and components of the processing platform **400**.
(33) The support circuits **498** are coupled to the CPU **496** for supporting the processor in a conventional manner. These circuits include cache, power supplies, clock circuits, input/output circuitry and subsystems, and the like. One or more processes may be stored in the memory **497** as software routine that, when executed or invoked by the processor, causes the processor to control the operation of the processing platform **400** or individual processing chambers in the manner described herein. The software routine may also be stored and/or executed by a second CPU (not shown) that is remotely located from the hardware being controlled by the CPU **496**.
(34) Some or all of the processes and methods of the present disclosure may also be performed in hardware. As such, the process may be implemented in software and executed using a computer system, in hardware as, e.g., an application specific integrated circuit or other type of hardware implementation, or as a combination of software and hardware. The software routine, when executed by the processor, transforms the general-purpose computer into a specific purpose computer (controller) that controls the chamber operation such that the processes are performed.
(35) In some embodiments, the controller **495** has one or more configurations to execute individual processes or sub-processes to perform the method. The controller **495** can be connected to and configured to operate intermediate components to perform the functions of the methods. For example, the controller **495** can be connected to and configured to control one or more of gas valves, actuators, motors, slit valves, vacuum control, or other components.
(36) FIGS. **5** through **6**D illustrate partial cross-sectional views of a substrate **500** (e.g., a semiconductor substrate **500**) with at least one feature **510** according to a method of one or more embodiments. The Figures show substrates having a single feature for illustrative purposes; however, those skilled in the art will understand that there can be more than one feature. The shape of the feature **510** can be any suitable shape including, but not limited to, trenches and cylindrical vias. As used in this regard, the term “feature” means any intentional surface irregularity. Suitable examples of features include, but are not limited to, trenches which have a top, two sidewalls and a bottom, peaks which have a top and two sidewalls. Features can have any suitable aspect ratio (ratio of the depth of the feature to the width of the feature). In some embodiments, the aspect ratio is greater than or equal to about 5:1, 10:1, 15:1, 20:1, 25:1, 30:1, 35:1, 40:1, 45:1, 50:1, 75:1, or 100:1.
(37) With reference to FIG. **5**, the substrate **500** has a substrate surface **520**. The at least one feature **510** forms an opening in the substrate surface **520**. The at least one feature **510** extends from the substrate surface **520** to a feature depth D.sub.f to a bottom surface **512**. The at least one feature **510** has a first sidewall **514** and a second sidewall **516** that define a width W of the at least one feature **510**. The open area formed by the sidewalls **514**, **516** and the bottom surface **512** are also referred to as a gap. In one or more embodiments, the width W is homogenous along the depth D.sub.1 of the at least one feature **510**. In other embodiments, the width, W, is greater at the top of the at least one feature **510** than the width, W, at the bottom surface **512** of the at least one feature **510**.
(38) In one or more unillustrated embodiments, the substrate **500** is a film stack comprising a

plurality of alternating layers of a first material and a second material. In some embodiments, the substrate **500** is a film stack comprising a plurality of alternating layers of a nitride material and an oxide material deposited on a semiconductor substrate. In other embodiments, the substrate **500** is a film stack comprising a plurality of alternating layers of silicon and silicon germanium on a semiconductor substrate.

(39) The semiconductor substrate **500** can be any suitable substrate material. In one or more embodiments, the semiconductor substrate **500** comprises a semiconductor material, e.g., silicon (Si), carbon (C), germanium (Ge), silicon germanium (SiGe), gallium arsenide (GaAs), indium phosphate (InP), indium gallium arsenide (InGaAs), indium aluminum arsenide (InAlAs), germanium (Ge), silicon germanium (SiGe), copper indium gallium selenide (CIGS), other semiconductor materials, or any combination thereof. In one or more embodiments, the semiconductor substrate **500** comprises one or more of silicon (Si), germanium (Ge), gallium (Ga), arsenic (As), indium (In), phosphorus (P), copper (Cu), or selenium (Se). Although a few examples of materials from which the substrate **500** may be formed are described herein, any material that may serve as a foundation upon which passive and active electronic devices (e.g., transistors, memories, capacitors, inductors, resistors, switches, integrated circuits, amplifiers, optoelectronic devices, or any other electronic devices) may be built falls within the spirit and scope of the present disclosure.

(40) In one or more embodiments, the at least one feature **510** comprises a memory hole or a word line slit. Accordingly, in one or more embodiments, the substrate **500** comprises a transistor, a memory device, or a logic device, e.g., FinFET, GAA, DRAM, VNAND, or the like.

(41) As used herein, the term “field effect transistor” or “FET” refers to a transistor that uses an electric field to control the electrical behavior of the device. Enhancement mode field effect transistors generally display very high input impedance at low temperatures. The conductivity between the drain and source terminals is controlled by an electric field in the device, which is generated by a voltage difference between the body and the gate of the device. The FET's three terminals are source (S), through which the carriers enter the channel; drain (D), through which the carriers leave the channel; and gate (G), the terminal that modulates the channel conductivity. Conventionally, current entering the channel at the source (S) is designated Is and current entering the channel at the drain (D) is designated I.sub.D. Drain-to-source voltage is designated V.sub.DS. By applying voltage to gate (G), the current entering the channel at the drain (i.e., I.sub.D) can be controlled.

(42) The metal-oxide-semiconductor field-effect transistor (MOSFET) is a type of field-effect transistor (FET). It has an insulated gate, whose voltage determines the conductivity of the device. This ability to change conductivity with the amount of applied voltage is used for amplifying or switching electronic signals. A MOSFET is based on the modulation of charge concentration by a metal-oxide-semiconductor (MOS) capacitance between a body electrode and a gate electrode located above the body and insulated from all other device regions by a gate dielectric layer. Compared to the MOS capacitor, the MOSFET includes two additional terminals (source and drain), each connected to individual highly doped regions that are separated by the body region. These regions can be either p or n type, but they are both be of the same type, and of opposite type to the body region. The source and drain (unlike the body) are highly doped as signified by a “+” sign after the type of doping.

(43) If the MOSFET is an n-channel or nMOS FET, then the source and drain are n+ regions and the body is a p region. If the MOSFET is a p-channel or pMOS FET, then the source and drain are p+ regions and the body is an n region. The source is so named because it is the source of the charge carriers (electrons for n-channel, holes for p-channel) that flow through the channel; similarly, the drain is where the charge carriers leave the channel.

(44) As used herein, the term “gate all-around (GAA),” is used to refer to an electronic device, e.g., a transistor, in which the gate material surrounds the channel region on all sides. The channel

region of a GAA transistor may include nanowires or nano-slabs or nano-sheets, bar-shaped channels, or other suitable channel configurations known to one of skill in the art. In one or more embodiments, the channel region of a GAA device has multiple horizontal nanowires or horizontal bars vertically spaced, making the GAA transistor a stacked horizontal gate-all-around (hGAA) transistor.

(45) As used herein, the term "nanowire" refers to a nanostructure, with a diameter on the order of a nanometer (10^{-9} meters). Nanowires can also be defined as the ratio of the length to width being greater than 1000. Alternatively, nanowires can be defined as structures having a thickness or diameter constrained to tens of nanometers or less and an unconstrained length. Nanowires are used in transistors and some laser applications, and, in one or more embodiments, are made of semiconducting materials, metallic materials, insulating materials, superconducting materials, or molecular materials. In one or more embodiments, nanowires are used in transistors for logic CPU, GPU, MPU, and volatile (e.g., DRAM) and non-volatile (e.g., VNAND) devices.

(46) As used herein, the term "dynamic random access memory" or "DRAM" refers to a memory cell that stores a datum bit by storing a packet of charge (i.e., a binary one), or no charge (i.e., a binary zero) on a capacitor. The charge is gated onto the capacitor via an access transistor and sensed by turning on the same transistor and looking at the voltage perturbation created by dumping the charge packet on the interconnect line on the transistor output. Thus, a single DRAM cell is made of one transistor and one capacitor. The DRAM device is formed of an array of DRAM cells. The rows on access transistors are linked by word lines, and the transistor inputs/outputs are linked by bit lines. Historically, DRAM capacitors have evolved from planar polysilicon-oxide-substrate plate capacitors to 3-D structures which have diverged into "stack" capacitors with both plates above the substrate, and "trench" capacitors using an etched cavity in the substrate as the common plate.

(47) Current DRAM buried word line (bWL) processes involve titanium nitride (TiN) and tungsten (W) stacks. Due to the poor adhesion between metal and trench structures, however, voiding and delamination of the metal fill is often observed during high-temperature post anneal treatments. Such voids and delamination are undesired because it will cause problems for subsequent planarization or etching processes. Voids and delamination also contribute to an increase in stack resistance. Accordingly, embodiments of the present disclosure provide processes for making a stack in a DRAM buried word line (bWL) on a substrate which advantageously does not have a seam.

(48) Traditionally, DRAM cells have recessed high work-function metal structures in buried word line structure. In a DRAM device, a bit line is formed in a metal level situated above the substrate, while the word line is formed at the polysilicon gate level at the surface of the substrate. In a buried word line (bWL) device, a word line is buried below the surface of a semiconductor substrate using a metal as a gate electrode.

(49) FIGS. **6**A through **6**D show a cross-sectional schematic of a metal gate fill process in accordance with one or more embodiments of the disclosure. FIG. **7** illustrates a process flow diagram of a method **700** in accordance with one or more embodiments of the disclosure. With reference to FIGS. **5** through **6**D and FIG. **7**, in one or more embodiments, at least one feature **510** is formed on a substrate **500**. In some embodiments, the substrate **500** is provided for processing prior to operation **702**. As used in this regard, the term "provided" means that the substrate is placed into a position or environment for further processing. In one or more embodiments, the substrate **500** has at least one feature **510** already formed thereon. In other embodiments, at operation **702**, at least one feature **510** is formed on a substrate **500**. In one or more embodiments, the at least one feature extends a feature depth, D_f, from the substrate surface **520** to a bottom surface, the at least one feature having a width, W, defined by a first sidewall **514** and a second sidewall **516**.

(50) In one or more embodiments, at operation **704**, a first film **530** is formed on the substrate

surface **520** and the sidewalls **514**, **516** and the bottom of the at least one feature **510**. As illustrated in FIG. **6**A, in one or more embodiments, the first film **530** has a void **540** located within the width, W, of the at least one feature **510**.

(51) In one or more embodiments, the first film **530** can be comprised of any suitable material. In some embodiments, the first film **530** comprises a metal nitride. In one or more embodiments, the metal nitride comprises one or more of titanium nitride (TiN), tantalum nitride (TaN), tungsten nitride (WN), silicon nitride (SiN), aluminum nitride (AlN), and the like. In one or more embodiments, the first film **530** is formed by atomic layer deposition or chemical vapor deposition.

(52) “Atomic layer deposition” or “cyclical deposition” as used herein refers to the sequential exposure of two or more reactive compounds to deposit a layer of material on a substrate surface. The substrate, or portion of the substrate, is exposed sequentially or separately to the two or more reactive compounds which are introduced into a reaction zone of a processing chamber. In a time-domain ALD process, exposure to each reactive compound is separated by a time delay to allow each compound to adhere and/or react on the substrate surface and then be purged from the processing chamber. These reactive compounds are said to be exposed to the substrate sequentially.

(53) In a spatial ALD process, different portions of the substrate surface, or material on the substrate surface, are exposed simultaneously to the two or more reactive compounds so that any given point on the substrate is substantially not exposed to more than one reactive compound simultaneously. As used in this specification and the appended claims, the term “substantially” used in this respect means, as will be understood by those skilled in the art, that there is the possibility that a small portion of the substrate may be exposed to multiple reactive gases simultaneously due to diffusion, and that the simultaneous exposure is unintended.

(54) In one aspect of a time-domain ALD process, a first reactive gas (i.e., a first precursor or compound A, e.g., manganese precursor, ruthenium precursor, or a manganese-ruthenium precursor) is pulsed into the reaction zone followed by a first time delay. Next, a second precursor or compound B (e.g., reductant) is pulsed into the reaction zone followed by a second delay. During each time delay, a purge gas, such as argon, may be introduced into the processing chamber to purge the reaction zone or otherwise remove any residual reactive compound or reaction by-products from the reaction zone. Alternatively, the purge gas may flow continuously throughout the deposition process so that only the purge gas flows during the time delay between pulses of reactive compounds. The reactive compounds are alternatively pulsed until a desired film or film thickness is formed on the substrate surface. In either scenario, the ALD process of pulsing compound A, purge gas, compound B, and purge gas is a cycle. A cycle can start with either compound A or compound B and continue the respective order of the cycle until achieving a film with the predetermined thickness.

(55) A “pulse” or “dose” as used herein is intended to refer to a quantity of a source gas that is intermittently or non-continuously introduced into the process chamber. The quantity of a particular compound within each pulse may vary over time, depending on the duration of the pulse. A particular process gas may include a single compound or a mixture/combination of two or more compounds, for example, the process gases described below.

(56) The durations for each pulse/dose are variable and may be adjusted to accommodate, for example, the volume capacity of the processing chamber as well as the capabilities of a vacuum system coupled thereto. Additionally, the dose time of a process gas may vary according to the flow rate of the process gas, the temperature of the process gas, the type of control valve, the type of process chamber employed, as well as the ability of the components of the process gas to adsorb onto the substrate surface. Dose times may also vary based upon the type of layer being formed and the geometry of the device being formed. A dose time should be long enough to provide a volume of compound sufficient to adsorb/chemisorb onto substantially the entire surface of the substrate and form a layer of a process gas component thereon.

(57) The precursor-containing process gas may be provided in one or more pulses or continuously.

The flow rate of the precursor-containing process gas can be any suitable flow rate including, but not limited to, flow rates is in the range of about 1 to about 5000 sccm, or in the range of about 2 to about 4000 sccm, or in the range of about 3 to about 3000 sccm or in the range of about 5 to about 2000 sccm. The precursor can be provided at any suitable pressure including, but not limited to, a pressure in the range of about 5 mTorr to about 500 Torr, or in the range of about 100 mTorr to about 500 Torr, or in the range of about 5 Torr to about 500 Torr, or in the range of about 50 mTorr to about 500 Torr, or in the range of about 100 mTorr to about 500 Torr, or in the range of about 200 mTorr to about 500 Torr.
(58) The period of time that the substrate is exposed to the one or more precursor-containing process gas may be any suitable amount of time necessary to allow the precursor to form an adequate nucleation layer atop the conductive surface of the bottom of the opening. For example, the process gas may be flowed into the process chamber for a period of about 0.1 seconds to about 90 seconds. In some time-domain ALD processes, the precursor-containing process gas is exposed the substrate surface for a time in the range of about 0.1 sec to about 90 sec, or in the range of about 0.5 sec to about 60 sec, or in the range of about 1 sec to about 30 sec, or in the range of about 2 sec to about 25 sec, or in the range of about 3 sec to about 20 sec, or in the range of about 4 sec to about 15 sec, or in the range of about 5 sec to about 10 sec.
(59) In some embodiments, an inert carrier gas may additionally be provided to the process chamber at the same time as the precursor-containing process gas. The carrier gas may be mixed with the precursor-containing process gas (e.g., as a diluent gas) or separately and can be pulsed or of a constant flow. In some embodiments, the carrier gas is flowed into the processing chamber at a constant flow in the range of about 1 to about 10000 sccm. The carrier gas may be any inert gas, for example, such as argon, helium, neon, combinations thereof, or the like. In one or more embodiments, a precursor-containing process gas is mixed with argon prior to flowing into the process chamber.
(60) In an embodiment of a spatial ALD process, a first reactive gas and second reactive gas (e.g., nitrogen gas) are delivered simultaneously to the reaction zone but are separated by an inert gas curtain and/or a vacuum curtain. The substrate is moved relative to the gas delivery apparatus so that any given point on the substrate is exposed to the first reactive gas and the second reactive gas.
(61) Suitable precursors include, but are not limited to, silane, disilane, dichlorosilane (DCS), trisilane, tetrasilane, titanium chloride, tantalum chloride, tungsten chloride, aluminum chloride, titanium fluoride, tantalum fluoride, tungsten fluoride, aluminum fluoride, and the like. In one or more embodiments, the precursor may be heated in a hot can to increase the vapor pressure and be delivered to the chamber using a carrier gas (e.g., ultrahigh purity (UHP) Ar, He, H.sub.2, etc.).
(62) In some embodiments, the first film **530** forms conformally on the at least one feature **510**. As used herein, the term “conformal”, or “conformally”, refers to a layer that adheres to and uniformly covers exposed surfaces with a thickness having a variation of less than 1% relative to the average thickness of the film. For example, a 1,000 Å thick film would have less than 10 Å variations in thickness. This thickness and variation includes edges, corners, sides, and the bottom of recesses. For example, a conformal layer deposited by ALD in various embodiments of the disclosure would provide coverage over the deposited region of essentially uniform thickness on complex surfaces.
(63) In some embodiments, the first film **530** is a continuous film. As used herein, the term “continuous” refers to a layer that covers an entire exposed surface without gaps or bare spots that reveal material underlying the deposited layer. A continuous layer may have gaps or bare spots with a surface area less than about 1% of the total surface area of the film.
(64) With reference to FIG. **6**A, in one or more embodiments, a void **540** may be formed within the width, W, of the at least one feature **510**. The shape and size of the void **540** can vary.
(65) Referring to FIG. **6**A and FIG. **7**, in one or more embodiments, at operation **706**, the film is treated with an oxidizing plasma to oxidize a portion of the first film **530**, forming an oxide film **550**. The plasma may comprise any suitable oxidizing plasma known to the skilled artisan. In one

or more embodiments, the oxidizing plasma comprises one or more of oxygen (O.sub.2), nitrous oxide (N.sub.2O), water (H.sub.2O), ozone (O.sub.3), and the like.
(66) In some embodiments, about 10% of the first film **530** is converted to an oxide film **550**. In other embodiments, about 20% or about 30% or about 50% or about 60% or about 70% of the first film **530** form an oxide film **550**. In one or more embodiments, the oxide film **550** comprises an oxidized metal nitride film. In one or more embodiments, the oxide film **550** comprises one or more of titanium oxynitride (TiON), tantalum oxynitride (TaON), tungsten oxynitride (WON), silicon oxynitride (SiON), aluminum oxynitride (AlON), and the like.
(67) Referring to FIG. **6**C and FIG. **9**, at operation **708**, the substrate **500** is etched to remove the oxide film **550** and expose the portion of first film **530** that was not oxidized by the oxidating plasma.
(68) The substrate may be etched, the oxide film **550** selectively removed, by any process known to one of skill in the art, including, but not limited to, wet etching, plasma-based sputter etching, chemical etching, Siconi® etching, reactive ion etching (RIE), high density plasma (HDP) etching, and the like. In some embodiments, etching the comprises exposing the substrate to an etch chemistry comprising a halide. In one or more embodiments, the oxide film **550** is etched by a dry etch halide process.
(69) Referring to FIG. **6**D and FIG. **7**, in one or more embodiments, at operation **710**, a second film **532** is deposited in the at least one feature **510** and on the substrate surface **520**.
(70) In one or more embodiments, the second film **532** can comprise any suitable material. In some embodiments, the second film **532** comprises a metal nitride film. In one or more embodiments, the metal nitride comprises one or more of titanium nitride (TiN), tantalum nitride (TaN), tungsten nitride (WN), silicon nitride (SiN), aluminum nitride (AlN), and the like.
(71) In one or more embodiments, the second film **532** is formed by atomic layer deposition or chemical vapor deposition. In one or more embodiments, the first film **530** and the second film **532** comprise the same material. In other embodiments, the first film **530** and the second film **532** comprise different materials.
(72) Referring to FIG. **7**, after the completion of process cycle **720**, decision **712** is reached. After a suitable number of sequences or cycles of deposition and treatment and etch, the metal nitride film formed will have substantially no seam to completely fill the gap of the feature **510**, as illustrated in FIG. **6**D. As used in this specification and the appended claims, the term “substantially no seam”, and the like, means that a seam takes up less than about 1% of the volume of the feature **510**. Referring to FIG. **6**D and FIG. **7**, substantially no seam is present. Thus, decision **712** would direct the process out of the deposition/treatment/etch loop to one or more optional post-processing treatments **714**.
(73) In some embodiments, operations **704**, **706**, **708**, and **710** are integrated such that there is no vacuum break between each operation. In some embodiments, operations **704** and **706** are integrated such that there is no vacuum break. In some embodiments, operations **704**. **706**, and **708** are integrated such that there is no vacuum break.
(74) According to one or more embodiments, the substrate is subjected to processing prior to and/or after forming the layer. This processing can be performed in the same chamber or in one or more separate processing chambers.
(75) Some embodiments of the disclosure are directed to film deposition processes using a batch processing chamber, also referred to as a spatial processing chamber. In one or more embodiments, the batch processing chamber may be any batch processing chamber known to one of skill in the art. In one or more embodiments, the deposition/treatment/etching/deposition occurs in the same processing tool. In one or more embodiments, the process zones in batch processing chamber are plasma etch capable.
(76) Several well-known cluster tools which may be adapted for the present disclosure are the Olympia®, the Continuum®, and the Trillium®, all available from Applied Materials, Inc., of

Santa Clara, Calif. However, the exact arrangement and combination of chambers may be altered for purposes of performing specific steps of a process as described herein. Other processing chambers which may be used include, but are not limited to, cyclical layer deposition (CLD), atomic layer deposition (ALD), chemical vapor deposition (CVD), physical vapor deposition (PVD), plasma treatment, etch, pre-clean, chemical clean, thermal treatment such as RTP, plasma nitridation, degas, hydroxylation and other substrate processes. By carrying out processes in a chamber on a cluster tool, surface contamination of the substrate with atmospheric impurities can be avoided without oxidation prior to depositing a subsequent film.

(77) According to one or more embodiments, the substrate is continuously under vacuum or “load lock” conditions and is not exposed to ambient air when being moved from one chamber to the next. The transfer chambers are thus under vacuum and are “pumped down” under vacuum pressure. Inert gases may be present in the processing chambers or the transfer chambers. In some embodiments, an inert gas is used as a purge gas to remove some or all of the reactants (e.g., reactant). According to one or more embodiments, a purge gas is injected at the exit of the deposition chamber to prevent reactants (e.g., reactant) from moving from the deposition chamber to the transfer chamber and/or additional processing chamber. Thus, the flow of inert gas forms a curtain at the exit of the chamber.

(78) The substrate can be processed in single substrate deposition chambers, where a single substrate is loaded, processed, and unloaded before another substrate is processed. The substrate can also be processed in a continuous manner, similar to a conveyer system, in which multiple substrates are individually loaded into a first part of the chamber, move through the chamber, and are unloaded from a second part of the chamber. The shape of the chamber and associated conveyer system can form a straight path or curved path. Additionally, the processing chamber may be a carousel in which multiple substrates are moved about a central axis and are exposed to deposition, etch, annealing, cleaning, etc. processes throughout the carousel path.

(79) During processing, the substrate can be heated or cooled. Such heating or cooling can be accomplished by any suitable means including, but not limited to, changing the temperature of the substrate support, and flowing heated or cooled gases to the substrate surface. In some embodiments, the substrate support includes a heater/cooler which can be controlled to change the substrate temperature conductively. In one or more embodiments, the gases (either reactive gases or inert gases) being employed are heated or cooled to locally change the substrate temperature. In some embodiments, a heater/cooler is positioned within the chamber adjacent the substrate surface to convectively change the substrate temperature.

(80) The substrate can also be stationary or rotated during processing. A rotating substrate can be rotated (about the substrate axis) continuously or in discrete steps. For example, a substrate may be rotated throughout the entire process, or the substrate can be rotated by a small amount between exposures to different reactive or purge gases. Rotating the substrate during processing (either continuously or in steps) may help produce a more uniform deposition or etch by minimizing the effect of, for example, local variability in gas flow geometries.

(81) Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below”, or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the exemplary term “below” may encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

(82) The use of the terms “a” and “an” and “the” and similar referents in the context of describing the materials and methods discussed herein (especially in the context of the following claims) are

to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., "such as") provided herein, is intended merely to better illuminate the materials and methods and does not pose a limitation on the scope unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the disclosed materials and methods.
(83) Reference throughout this specification to "one embodiment," "certain embodiments," "one or more embodiments" or "an embodiment" means that a particular feature, structure, material, or characteristic described in connection with the embodiment is included in at least one embodiment of the disclosure. Thus, the appearances of the phrases such as "in one or more embodiments," "in certain embodiments," "in one embodiment" or "in an embodiment" in various places throughout this specification are not necessarily referring to the same embodiment of the disclosure. In one or more embodiments, the particular features, structures, materials, or characteristics are combined in any suitable manner.
(84) Although the disclosure herein has been described with reference to particular embodiments, it is to be understood that these embodiments are merely illustrative of the principles and applications of the present disclosure. It will be apparent to those skilled in the art that various modifications and variations can be made to the method and apparatus of the present disclosure without departing from the spirit and scope of the disclosure. Thus, it is intended that the present disclosure include modifications and variations that are within the scope of the appended claims and their equivalents.

Claims

1. A processing method comprising: depositing a first film on a substrate surface, the substrate surface having at least one feature thereon, the at least one feature extending a feature depth from the substrate surface to a bottom surface, the at least one feature having a width defined by a first sidewall and a second sidewall, the first film forming conformally on the substrate surface, and on the first sidewall, the second sidewall, and the bottom surface of the at least one feature, and having a void located within the width of the at least one feature; treating the substrate surface with an oxidizing plasma to oxidize a portion of the first film formed on the substrate surface, and on the first sidewall, the second sidewall, and the bottom surface of the at least one feature, and form an oxidized film; etching the substrate surface to remove the oxidized film; and depositing a second film on the substrate surface to fill the at least one feature, the second film substantially free of seams and voids.
2. The method of claim 1, wherein the first film and the second film independently comprise a metal nitride.
3. The method of claim 2, wherein the metal nitride comprises one or more of titanium nitride (TiN), tantalum nitride (TaN), tungsten nitride (WN), silicon nitride (SiN), and aluminum nitride (AlN).
4. The method of claim 3, wherein the oxidized film comprises one or more of titanium oxynitride (TiON), tantalum oxynitride (TaON), tungsten oxynitride (WON), silicon oxynitride (SiON), and aluminum oxynitride (AlON).
5. The method of claim 1, wherein depositing the first film, and depositing the second film comprises an atomic layer deposition process.
6. The method of claim 1, further comprising repeating the method n number of times, wherein n is a number less than or equal to 40 cycles.

7. The method of claim 1, wherein the feature has an aspect ratio greater than or equal to about 10:1.
8. The method of claim 1, wherein the oxidizing plasma comprises one or more of oxygen (O.sub.2), nitrous oxide (N.sub.2O), water (H.sub.2O), and ozone (O.sub.3).
9. The method of claim 1, wherein about 10% to about 70% of the first film is converted to the oxidized film.
10. A processing method comprising: forming a film stack on a substrate, the film stack comprising a plurality of alternating layers of a first material and a second material and the film stack having a stack thickness; etching the film stack to form an opening extending a depth from a top of the film stack surface to a bottom surface, the opening having a width defined by a first sidewall and a second sidewall; depositing a first film on the film stack surface, the first film forming conformally on the film stack surface, and on the first sidewall, the second sidewall, and the bottom surface of the opening, and having a void located within the width of the opening; treating the film stack with an oxidizing plasma to oxidize a portion and form an oxidized film on the film stack surface, and on the first sidewall, the second sidewall, and the bottom surface of the opening; etching the oxidized film to expose the first film; and depositing a second film on the film stack surface to fill the opening, the second film substantially free of seams and voids.
11. The method of claim 10, wherein the first film and the second film independently comprise a metal nitride.
12. The method of claim 11, wherein the metal nitride comprises one or more of titanium nitride (TiN), tantalum nitride (TaN), tungsten nitride (WN), silicon nitride (SiN), and aluminum nitride (AlN).
13. The method of claim 12, wherein the oxidized film comprises one or more of titanium oxynitride (TION), tantalum oxynitride (TaON), tungsten oxynitride (WON), silicon oxynitride (SiON), and aluminum oxynitride (AION).
14. The method of claim 10, wherein depositing the first film, and depositing the second film comprises an atomic layer deposition process.
15. The method of claim 10, further comprising repeating the method n number of times, wherein n is a number less than or equal to 40 cycles.
16. The method of claim 10, wherein the first material and the second material independently comprise one or more of silicon oxide (SiOx), silicon nitride (SiN), silicon (Si), and silicon germanium (SiGe).
17. The method of claim 10, wherein the oxidizing plasma comprises one or more of oxygen (O.sub.2), nitrous oxide (N.sub.2O), water (H.sub.2O), and ozone (O.sub.3).
18. The method of claim 10, wherein about 10% to about 70% of the first film is converted to the oxidized film.
19. A non-transitory computer readable medium including instructions, that, when executed by a controller of a processing chamber, causes the processing chamber to perform the operations of: deposit a first film on a substrate surface, the substrate surface having at least one feature thereon, the at least one feature extending a feature depth from the substrate surface to a bottom surface, the at least one feature having a width defined by a first sidewall and a second sidewall, the first film forming conformally on the substrate surface, and on the first sidewall, the second sidewall, and the bottom surface of the at least one feature, and having a void located within the width of the at least one feature; treat the substrate surface with an oxidizing plasma to form an oxidized film from the conformal first film on the substrate surface, and on the first sidewall, the second sidewall, and the bottom surface of the at least one feature; etch the substrate surface to remove the oxidized film; and deposit a second film on the substrate surface to fill the at least one feature, the second film substantially free of seams and voids.
20. The non-transitory computer readable medium of claim 19, further including instructions, that, when executed by a controller of a processing chamber, causes the processing chamber to perform

the further operation of: repeat the operations of depositing the first film on the substrate surface, treating the substrate surface with the oxidizing plasma, etching the substrate surface, and depositing the second film, n number of times, wherein n is a number less than or equal to 40 cycles.